

Event Information:

EventID :
44
Event Time :
Jan, 31, 2021, 04:59 PM
Rule :
SOC113 - Suspicious hh.exe Usage
Level :
Security Analyst
Source Address :
172.16.17.47
Source Hostname :
BillPRD
File Name :
WinRAR.chm
File Hash :
07694464c25bac4ecdb365e928ffe1ff
File Size :
306 KB
Device Action :
Allowed

Analysis Steps:

Email and network logs were searched with IP 172[.]16.17.47 (BillPRD) as the source in the relevant time-frame but no events were found that would indicate how the file got to the device.

When searching the MD5 hash on VirusTotal it is listed as clean.

The screenshot shows the VirusTotal interface for a file named 'WinRAR.chm'. The file size is 306.95 KB and it was last analyzed 19 days ago. The detection is 'detect-debug-environment'. A table shows security vendors' analysis results, all of which are 'Undetected'.

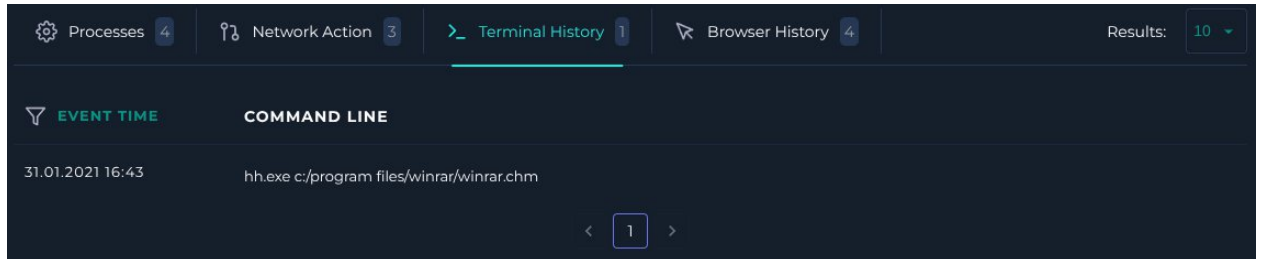
Security vendors' analysis	Do you want to automate checks?
Acronis (Static ML)	Undetected
AlCloud	Undetected
Antiy-AVL	Undetected
Avast	Undetected

Multiple tests on AnyRun also determine the file to be safe.

The screenshot shows the AnyRun interface with test results for 'WinRAR.chm'. The file is safe across multiple operating systems and architectures. The results are as follows:

OS	Architecture	Test Result	File Name	MD5	SHA1	SHA256
Windows 10 Professional 64 bit	64 bit	No threats detected	WinRAR.chm	07694464c25bac4ecdb365e928ffe1ff	EBE42D5830AE5A4A21639CEE011D0B67B93433E	D7E601BCE0987973F76F6CD6FB49A011B4FB86EA060196C7CF2EC21BB9...
Windows 10 Professional 64 bit	64 bit	No threats detected	WinRAR.chm	07694464c25bac4ecdb365e928ffe1ff	EBE42D5830AE5A4A21639CEE011D0B67B93433E	D7E601BCE0987973F76F6CD6FB49A011B4FB86EA060196C7CF2EC21BB9...
Windows 10 Professional 64 bit	64 bit	No threats detected	07694464c25bac4ecdb365e928ffe1ff.zip	4E3BA2A463A1704A102219A92896D92	D819A40F71F33C6B26A0FCEFC5189961DF6843D	5F08A68CF63F28D4FDE2A48CD39EC15DDBC524EDA58AC40BB172002D333...
Windows 7 Professional 32 bit	32 bit	No threats detected	WinRAR.chm	07694464c25bac4ecdb365e928ffe1ff	EBE42D5830AE5A4A21639CEE011D0B67B93433E	D7E601BCE0987973F76F6CD6FB49A011B4FB86EA060196C7CF2EC21BB9...

hh.exe is the standard, built-in Windows HTML Help executable. Its legitimate function is simply to open and display .chm (Compiled HTML) help manuals. The terminal history shows the exact command hh.exe c:/program files/winrar/winrar.chm was executed. The user most likely clicked the "Help" button or pressed the F1 key while using the WinRAR application. This normal action prompted Windows to use hh.exe to open the official WinRAR instruction manual located in its default installation folder.



The screenshot shows a security tool interface with a dark theme. At the top, there are four tabs: 'Processes' (4 items), 'Network Action' (3 items), 'Terminal History' (1 item, which is selected and highlighted with a red underline), and 'Browser History' (4 items). On the far right, it says 'Results: 10' with a dropdown arrow. Below the tabs is a table with two columns: 'EVENT TIME' and 'COMMAND LINE'. The table contains one row with the event time '31.01.2021 16:43' and the command line 'hh.exe c:/program files/winrar/winrar.chm'. At the bottom right of the table, there is a pagination control showing '< 1 >'.

EVENT TIME	COMMAND LINE
31.01.2021 16:43	hh.exe c:/program files/winrar/winrar.chm

The reason this flagged an alert is that hh.exe is often abused by threat actors as a LOL (Living off the Land) binary, as a means of detection avoidance. In this scenario, however, all evidence leads to this being a False Positive.

Based on following Playbook procedure and the analysis of available information, this alert is determined to be a False Positive.

Report:

Incident Summary

Analysis confirms a False Positive alert for suspicious hh.exe usage on the host BillPRD (172[.]16.17.47). The alert was triggered by the standard Windows HTML Help executable (hh.exe) opening a legitimate compiled help file (WinRAR.chm). Threat actors frequently abuse hh.exe as a Living off the Land (LOL) binary to avoid detection, which is why the SIEM generated this alert. However, telemetry and sandbox testing confirm this specific event was benign user activity.

Incident Details

- **Alert Name:** SOC113 - Suspicious hh.exe Usage
- **Target Host:** BillPRD (172[.]16.17.47)
- **Flagged File:** WinRAR.chm
- **File Hash:** 07694464c25bac4ecdb365e928ffe1ff
- **Device Action:** Allowed

Analysis and Evidence

- **Execution:** Terminal history confirms the command hh.exe c:/program files/winrar/winrar.chm was executed on January 31, 2021, at 16:43.
- **Root Cause:** This execution pattern indicates the user likely pressed the F1 key or clicked the Help button within the WinRAR application. This prompted Windows to open the official instruction manual from its default installation directory.
- **Delivery Mechanism:** Searches across email and network logs for the relevant time frame found no events indicating malicious delivery of the file to the endpoint.

Threat Intelligence

- **OSINT:** VirusTotal analysis of the file hash (07694464c25bac4ecdb365e928ffe1ff) returned a 0/60 clean score.
- **Sandbox Testing:** Multiple dynamic analysis tests on AnyRun determined the WinRAR.chm file to be completely safe, returning a "No threats detected" result.

Actions Taken

1. Verified the file hash against OSINT sources (VirusTotal), confirming it is clean.
2. Reviewed sandbox telemetry (AnyRun), noting the file exhibits no malicious behaviour.
3. Analysed endpoint terminal history, confirming the execution was a legitimate request to open the WinRAR help manual.
4. Determined the alert to be a False Positive triggered by a known LOLBin rule and closed the ticket.

Let's Defend Result

SOC113 - Suspicious hh.exe Usage

False Positive (+5 Point)

Analyze Malware (+5 Point)

Check if the malware is quarantined/cleaned (+5 Point)